

In-season estimates of ad-intact fish returning to the Krassel PIT-array

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October 18, 2023

Summary

July 17, 2023

- *The estimated total ad-intact fish at KRS was 1,414 (949 NOR + 465 Integrated)*
- Natural-origin
 - 656 natural-origin fish were estimated based on 118 observed tags and a 18% sample/tag rate at Lower Granite Dam.
 - KRS was non-operational: July 10-11 and July 14-16. Estimated natural-origin PIT-tags passing KRS during the non-operational period was estimated as ≤ 10 tags for an additional 56 natural-origin fish.
 - The average passage date of $\geq 75\%$ of the total return across KRS is July 15th (Figure 1).
 - End of season natural-origin estimate predicted as 949.
 - NOR: $(656 + 56)/0.75 = 949$.
- Integrated
 - Estimated 465 McCall Integrated fish at Lower Granite Dam.
 - Estimated 443 at KRS after accounting for SbyC (i.e., 95% conversion).

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- *Current estimated total ad-intact fish at KRS was 1,439 (912 NOR + 527 Integrated)*¹
- Natural-origin
 - 856 natural-origin fish estimated based on 154 observed tags and a 18% sample/tag rate at Lower Granite Dam.
 - Assume 100% passage past KRS.

¹Scripts to reproduce these summaries are available at: “https://github.com/NPTfisheries/sf_salmon_harvest.com”.

- End of season natural-origin fish estimated as 856.
- NOR: $(856 + 56)/1.00 = 912$.
- Integrated
 - Estimate 527 McCall Integrated fish at KRS based on 108 tags observed and “group-specific” expansion rates (Tables 1 and 2)

Assumptions

- 100% conversion of integrated fish to KRS.
- 100% detection efficiency at KRS
 - In 2021 and 2022 KRS detections efficiency was $\geq 95\%$.
- Tagging rate at LGR of 18%.
 - Only 4 out of 118 NOR tags were marked at LGR during the 29% sampling rate, all others marked at 18%.
- Static interpolation rate of 2 tags during the missed period at KRS when observations are typically decreasing.

Table 1: *Current tag expansion rates and estimates for SFSR release groups at KRS.*

Release Year	Group	SbyC	Expansion	n_tags	Estimate
2020	Integrated	RAL	6.710924	1	7
2020	Integrated	RTR	1.000000	1	1
2020	Segregated	RAL	50.028692	1	50
2021	Integrated	RAL	6.995739	56	392
2021	Integrated	RTR	1.000000	24	24
2021	Segregated	RAL	50.761811	50	2,538
2021	Segregated	RTR	1.000000	28	28
2022	Integrated	RAL	8.312441	12	100
2022	Integrated	RTR	1.000000	3	3
2022	Integrated			4	
2022	Segregated	RAL	39.299292	10	393
2022	Segregated	RTR	1.000000	5	5

Table 1: *Current tag expansion rates and estimates for SF SR release groups at KRS.*

Release Year	Group	SbyC	Expansion	n_tags	Estimate
2022	Segregated			1	
2023	NOR		5.555556	154	856

Table 2: *Current SF SR release group estimates at KRS.*

Group	n_tags	Estimate
Integrated	101	527
NOR	154	856
Segregated	95	3,014

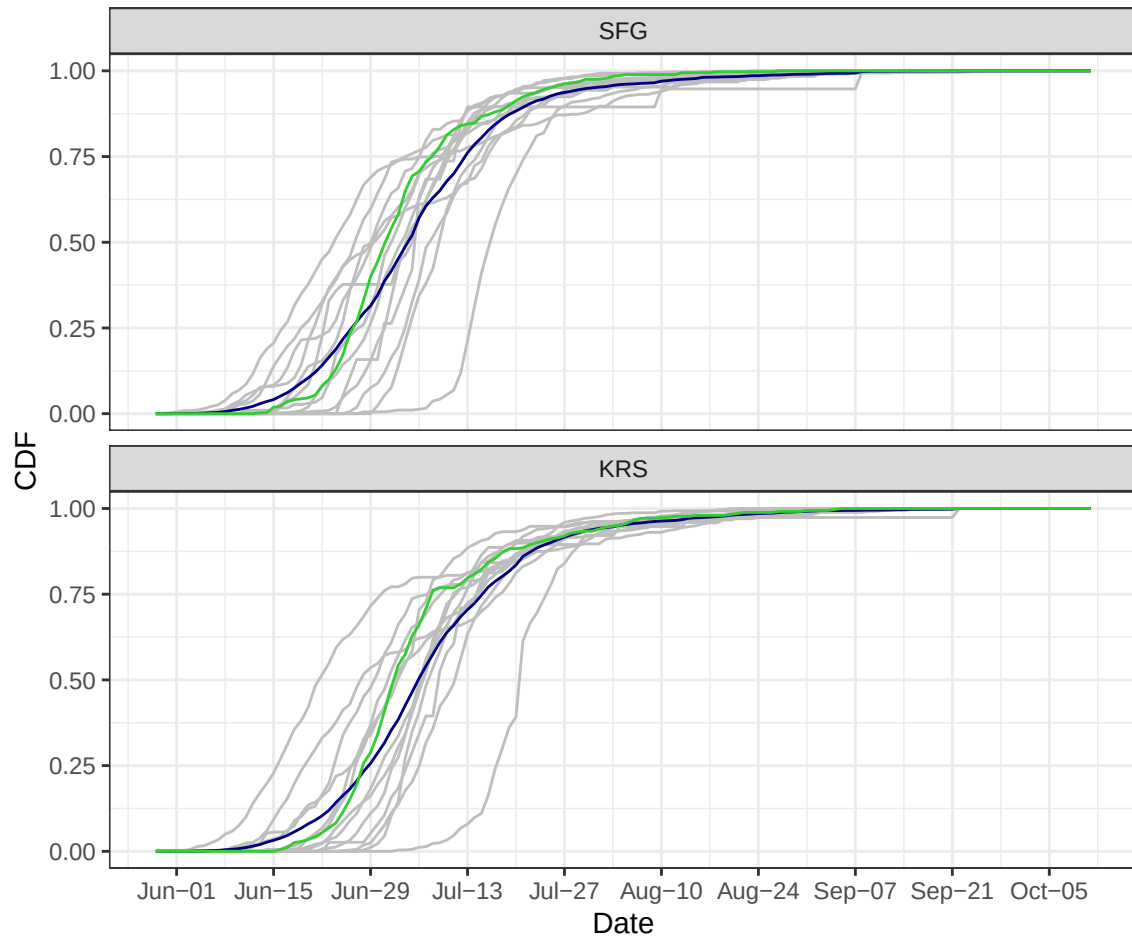


Figure 1: Run-timing of SFSR hatchery (segregated and integrated) and natural origin returns across PIT-arrays. The grey lines indicate individual spawn year returns for 2010-2022, the dark blue line is the average cumulative proportion of returns across the same years, and the green line represents the current 2023 return.